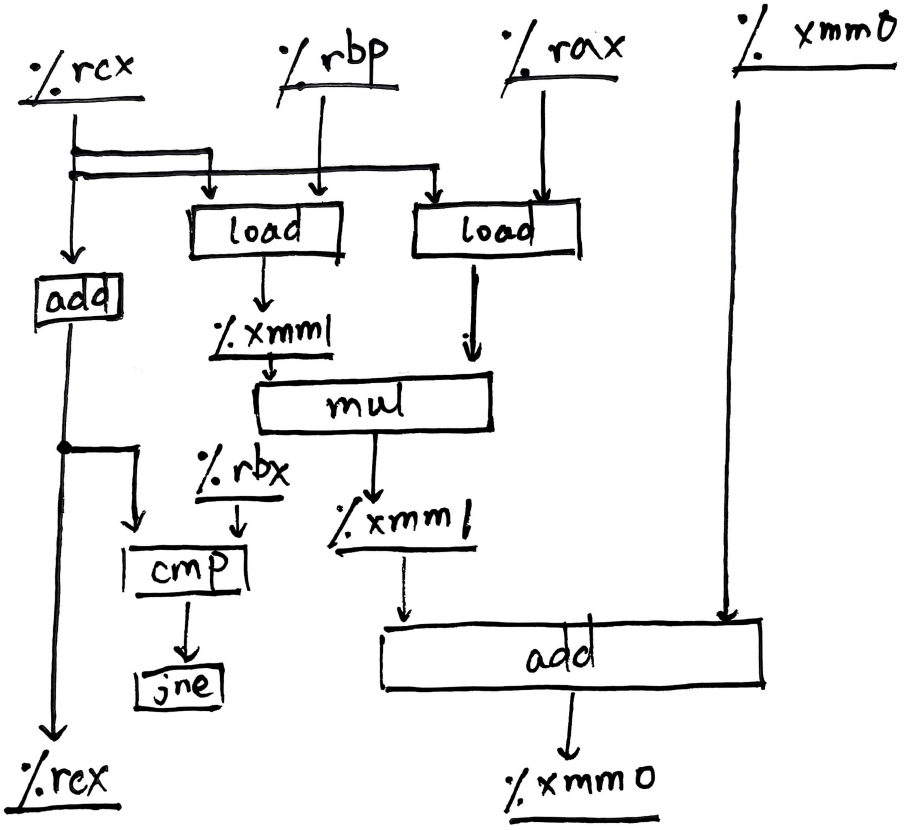
1. (P393 5.13-5.16)

(13)

(13-A)



关键路径: $\%\text{xmm0} \to \text{add} \to \%\text{xmm0} \to \text{add} \to \cdots$.

(13-B) fp add unit 的 latency bound.

(13-C) int add unit 的 latency bound.

(13-D) 乘法数据互不依赖, 可以流水化; 但加法数据前后依赖, 受 latency bound 限制.

(14)

```
void inner_6x1(vec_ptr u, vec_ptr v, data_t *dest) {
    long i, length = vec_length(u), limit = n - 5;
    data_t *udata = get_vec_start(u), *vdata = get_vec_start(v);
    data_t sum = 0;
    for (i = 0; i < limit; i += 6) {
        sum = sum + udata[i] * vdata[i] + udata[i + 1] * vdata[i + 1]
                  + udata[i + 2] * vdata[i + 2] + udata[i + 3] * vdata[i + 3]
                  + udata[i + 4] * vdata[i + 4] + udata[i + 5] * vdata[i + 5];
    }
    for (; i < length; i++) {
        sum = sum + udata[i] * vdata[i];
    }
    *dest = sum;
```

```
}
```

(14-A) load $2n$ 个整型或浮点型已经有 $\mathrm{CPE} \geq 1.00$.

(14-B) 所有浮点乘法间无数据依赖, 可以充分填充两个 pipelined fp mult units, 但是 `inner4` 的瓶颈在于浮点加法: 所有加法前后依赖, 所以由 fp add unit 的 latency bound 限制 CPE. `inner_6x1` 还是存在这一问题, 不能带来显著优化.

(15)

```c
void inner_6x6(vec_ptr u, vec_ptr v, data_t *dest) {
    long i, length = vec_length(u), limit = n - 5;
    data_t *udata = get_vec_start(u), *vdata = get_vec_start(v);
    data_t s0 = 0, s1 = 0, s2 = 0, s3 = 0, s4 = 0, s5 = 0;
    for (i = 0; i < limit; i += 6) {
        s0 = s0 + udata[i] * vdata[i];
        s1 = s1 + udata[i + 1] * vdata[i + 1];
        s2 = s2 + udata[i + 2] * vdata[i + 2];
        s3 = s3 + udata[i + 3] * vdata[i + 3];
        s4 = s4 + udata[i + 4] * vdata[i + 4];
        s5 = s5 + udata[i + 5] * vdata[i + 5];
    }
    for (; i < length; i++) {
        s0 = s0 + udata[i] * vdata[i];
    }
    *dest = s0 + s1 + s2 + s3 + s4 + s5;
}
```

制约因素: 关键路径外的 cycles, 过大的循环尾项常数等.

(16)

```c
void inner_6x1a(vec_ptr u, vec_ptr v, data_t *dest) {
    long i, length = vec_length(u), limit = n - 5;
    data_t *udata = get_vec_start(u), *vdata = get_vec_start(v);
    data_t sum = 0;
    for (i = 0; i < limit; i += 6) {
        sum = sum + ((udata[i] * vdata[i] + udata[i + 1] * vdata[i + 1])
                  +  (udata[i + 2] * vdata[i + 2] + udata[i + 3] * vdata[i + 3])
                  +  (udata[i + 4] * vdata[i + 4] + udata[i + 5] * vdata[i + 5]));
    }
    for (; i < length; i++) {
        sum = sum + udata[i] * vdata[i];
    }
    *dest = sum;
}
```

2. (P497 7.6)

符号	swap.o.symtab 条目?	符号类型	定义符号的模块	节
buf	是	外部	m.o	.data
bufp0	是	全局	swap.o	.data
bufp1	是	局部	swap.o	.bss
swap	是	全局	swap.o	.text
temp	否	/	/	/
incr	是	局部	swap.o	.text
count	是	局部	swap.o	.bss

3. (P498 7.8)

(A) $\mathrm{DEF}(\mathrm{main.1}), \mathrm{DEF}(\mathrm{main.2})$;

(B) 未知 (只有两个弱符号);

(C) 错误 (有两个强符号).

4. (P499 7.12)

(A) $4004\mathrm{f8} + (-4) - (4004\mathrm{e0} + \mathrm{a}) = \mathrm{a}$.

(B) $400500 + (-4) - (4004\mathrm{d0} + \mathrm{a}) = 22$.